

REVIEW ARTICLE

Incidence and risk factors of neonatal hypothermia: A systematic review and meta-analysis

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Abstract

Aim: Hypothermia poses a threat to the health and lives of newborns. Therefore, it is essential to identify the factors that influence neonatal hypothermia and provide targeted intervention suggestions for clinical practice to reduce its occurrence.

Methods: We conducted a literature search to identify factors influencing neonatal hypothermia and performed a meta-analysis to determine the prevalence of neonatal hypothermia and its associated factors. The Newcastle-Ottawa Scale (NOS) was used to assess the quality of cohort and case-control studies, while the Agency for Healthcare Research and Quality (AHRQ) was used to evaluate the quality of cross-sectional studies.

Results: Eighteen studies involving 44 532 newborns from 13 countries were included. The incidence of neonatal hypothermia was 52.5% (95% CI: 0.37, 0.68). Factors such as no skin-to-skin contact, prematurity, low birth weight, delayed breastfeeding, asphyxiation and resuscitation after birth, low APGAR score, not wearing a cap, and caesarean section were found to affect neonatal hypothermia.

Conclusion: Multiple factors influence neonatal hypothermia, and clinicians can utilise these factors to develop targeted intervention measures to prevent and reduce the incidence of neonatal hypothermia.

KEYWORDS

incidence rate, influence factor, meta-analysis, neonatal hypothermia

1 | INTRODUCTION

Newborns are easily affected by external environmental temperatures due to their underdeveloped thermoregulatory centre, large body surface area, and lack of brown adipose tissue, which can lead to hypothermia.^{1,2} The World Health Organisation (WHO)

defines hypothermia as a core body temperature below 36.5°C or a skin temperature below 36°C.¹ Hypothermia in neonates has a significant impact on their bodies, increasing the risk of infection; consequently, it is associated with pulmonary diseases, neonatal necrotising enterocolitis, acute renal failure, intracranial haemorrhage and even death.³ Studies have reported that

Abbreviations: AHRQ, Agency for Healthcare Research and Quality; APGAR score, A useful mnemonic to describe the components of the score: appearance, pulse, grimace, activity and respiration.; CBM, China Biology Medicine Literature; CI, Confidence intervals; CNKI, China National Knowledge Infrastructure; ELBW, Extremely low birth weight; ENC, Essential Newborn Care; NOS, Newcastle-Ottawa Scale; OR, Odds ratio; PRISMA, Preferred reporting items for systematic reviews of interventions; RR, Relative risk; VLBW, Very low birth weight; WHO, World Health Organisation.

hypothermia occurs in 32%–85% of neonates born in hospitals and approximately 11%–92% of those born at home.³ Hypothermia is significantly associated with neonatal death.^{4,5} Studies have demonstrated that the mortality rate of neonatal hypothermia increases by 28% for every 1°C decrease in neonatal temperature,⁶ and the global mortality rate of neonatal hypothermia ranges from 8.5% to 52%.³

For newborns, effective resuscitation, maintenance of body temperature, blood glucose levels and stability of the internal environment are important during the golden hour of life.⁷ The WHO recommends Essential Newborn Care (ENC), which includes interventions such as early initiation of breastfeeding, temperature regulation (drying/wrapping the baby, skin-to-skin contact and delayed bathing) and appropriate umbilical cord care.^{8,9} To achieve the United Nations Millennium Development Goals and reduce neonatal mortality, neonatal hypothermia must be promptly prevented.

Clinical practices include promoting and utilising various measures to prevent hypothermia.^{2,10} However, neonatal hypothermia continues to be a problem owing to variations in how its occurrence and clinical importance are perceived by medical professionals and family members in different regions. Additionally, disparities exist in the availability of resources at the regional level, including high-income countries and resource-poor areas.^{3,11–13}

A significant number of studies have reported the risk factors associated with neonatal hypothermia, but the findings are inconsistent. Delavar et al.¹⁴ found that neonatal hypothermia was primarily associated with temperatures lower than 27°C in delivery and postpartum rooms. However, they did not find a significant association with gestational age, delayed use of appropriate clothing and skin contact. In contrast, Bayih et al.¹⁵ and Girma et al.¹⁶ found that these factors were the primary influencing factors. Existing systematic reviews have only analysed the risk factors of neonatal hypothermia in low-income countries in East Africa¹⁷ and in very low or extremely low birth weight infants,¹⁸ while ignoring the current situation in other countries and in normal-weight infants. Therefore, this study systematically summarises the factors that influence neonatal hypothermia. The aims of this study were as follows:

1. Determine the current incidence of neonatal hypothermia.
2. Understanding the factors influencing neonatal hypothermia.

2 | METHODS

The study was registered on the PROSPERO website (<https://www.crd.york.ac.uk/PROSPERO/>) with registration number CRD42022381767 and followed the PRISMA guidelines.¹⁹

(1) The inclusion criteria stipulated that the subjects were infants aged 28 days or younger. (2) The content addressed risk factors associated with neonatal hypothermia. (3) The study design included cohort, case-control, or cross-sectional methodologies. (4) The data presented in the literature were extensive and appropriate for the computation of relative risk (RR) or odds ratio (OR) values, as well as

Key notes

- The incidence of neonatal hypothermia is high and poses a threat to the health and life of newborns.
- Prematurity, low birth weight, no skin-to-skin contact, delayed breastfeeding, asphyxia resuscitation after birth, low APGAR score, not wearing a cap and caesarean section can contribute to hypothermia in newborns.
- In the future, early intervention for neonatal hypothermia should be based on the influencing factors.

their respective 95% confidence intervals (CI). (5) Articles published in English or Chinese were included in the consideration process. The exclusion criteria were as follows: (1) case reports, conference abstracts, duplicate articles and interventions. (2) Studies with incomplete data or those lacking calculable odds ratio were excluded.

2.1 | Literature retrieval strategy

The English database search included PubMed, EMBASE, Web of Science, Cochrane Library and Chinese databases such as China Biology Medicine Literature (CBM), China National Knowledge Infrastructure (CNKI) and Wanfang, from the inception of the databases until 30 November 2022. English search terms included 'Infant*', 'Newborn*', 'Newborn Infant*', 'Neonate*', 'Hypothermia*', 'Accidental Hypothermia*', 'Temperature*', 'risk factors*', 'related factors*' and 'influencing factors*'. In addition, the references of the included studies were traced to obtain more relevant literature.

2.2 | Literature screening and data extraction

All studies were imported into EndNote and duplicate articles were excluded. The titles and abstracts were independently read by two researchers in order to eliminate irrelevant literature. Subsequently, the entire text was read and articles with results that did not align with the research theme were excluded based on the inclusion and exclusion criteria. Disagreements were resolved through mutual discussion or consultation with a third researcher (the corresponding author). Excel was used to extract the following information: title, author, publication time, country, research type, research topic, sample size, hypothermia determination, influencing factors, OR values and 95% CI values.

2.3 | Quality assessment of the included studies

Two researchers used the Newcastle-Ottawa Scale (NOS)²⁰ to evaluate the quality of the case-control and cohort studies. One star represents one point. The total score is nine points. Low-,

moderate- and high-quality studies scored 0–3, 4–6 and 7–9 points, respectively. Studies with more than five points were included in the meta-analysis. The quality of the cross-sectional study was evaluated using the Agency for Healthcare Research and Quality (AHRQ),²¹ a healthcare quality and research institute. The evaluation criteria of AHRQ included 11 items, all of which were assessed using 'yes', 'no' and 'unclear'. 'Yes' had a score of 1, while the rest were scored 0. The total score was categorised as follows: 0–3; low quality; 4–7; medium quality; and 8–11, high quality.

2.4 | Statistical analysis

Stata 14.0 was used for the meta-analysis. The effect indicators were analysed by calculating the odds ratios (ORs) and 95% confidence intervals (CIs). Heterogeneity between studies was evaluated using the Q test and I^2 statistic. Homogeneity was considered satisfactory when $p > 0.1$ and $I^2 \leq 50\%$. In this case, a fixed-effects model was chosen. Heterogeneity was considered high when $p \leq 0.1$ and $I^2 > 50\%$, and the random-effects model was selected. Both fixed-effects and random-effects models can be utilised to assess sensitivity between studies. Publication bias was quantitatively assessed using Egger's test, where $p > 0.05$ indicates no publication bias and $p < 0.05$ indicates the presence of publication bias.

3 | RESULTS

3.1 | Literature screening process and results

A total of 2112 articles were retrieved, including 190 from the Cochrane Library, 654 from PubMed, 600 from EMBASE, 216 from Web of Science, 118 from CNKI, 193 from Wanfang Data and 141 from CBM. A total of 876 duplicate articles were excluded from analysis. Furthermore, 1236 articles were excluded because of the inadequacy of their titles and abstracts. Additionally, 1115 articles deemed irrelevant to the study subjects were excluded after reviewing their titles and abstracts. After a thorough examination of the complete text, 102 articles were deemed ineligible for inclusion, leaving a total of 18 articles. The process of reviewing the literature and findings is illustrated in Figure 1.

3.2 | Characteristics and quality evaluation of the included studies

A total of 18 articles were included, comprising nine cross-sectional survey studies,^{14,15,22–28} three cohort studies^{12,13,29} and six case-control studies.^{16,30–34} The studies included 44 532 neonates from 13 countries, including Ethiopia, Iran, the Netherlands, Italy, the USA, Uganda, Nepal, South Africa, Canada, Kenya, Nigeria, China and Rwanda. The survey period was 2009–2022. The diagnostic criteria for hypothermia were an axillary temperature or skin

temperature below 36.5°C in 16 studies, an axillary temperature below 36°C in one study, and an axillary temperature below 35°C in one study. In eight studies, neonates had low birth weight and were premature infants. In five studies, neonates were admitted to the NICUs. Fifteen articles were of high quality, while three were of medium quality, indicating that the overall quality of the literature included in this study was high. The basic characteristics and quality of the included studies are summarised in Table 1.

3.3 | Meta-analysis results

This study included 31 factors that influenced outcomes. We conducted a meta-analysis of the influencing factors commonly described in more than three articles. We found that the following factors were associated with an increased risk of neonatal hypothermia: no skin-to-skin contact, prematurity, caesarean section, delayed breastfeeding, not wearing a cap, low birth weight, asphyxia resuscitation after birth and low Apgar score (Table 2).

We conducted a descriptive analysis of the influencing factors reported in fewer than three studies. These factors included no warm intra-facility transportation, mothers having obstetric complication, neonatal health problem, natural labour, night-time delivery, hypothermia at admission, analgesia administration, delayed cord clamping ($>60''$ vs. $<60''$), maternal temperature, room temperature, multiple gestation, race (Black), home birth, sex (Female), no thermal education, no at least 1KMC step, bathed within 24 h, place of birth (out), antenatal use of full course of steroid, maternal age, neonatal diseases occurred within 24 h after birth, admissions during cold season, age of Neonate <24 h old.

3.4 | Publication bias assessment

Publication bias was assessed using funnel plots and Egger's test for factors with five or more inclusions. The results indicated potential publication bias in the following factors: no skin-to-skin contact, prematurity, delayed breastfeeding, low birth weight, asphyxiation and resuscitation after birth. The results of Egger's test showed $p > |t|$ values of 0.005, 0.007, 0.003, 0.023 and 0.002, respectively ($p < 0.05$).

4 | DISCUSSION

This meta-analysis included 18 studies published up to 30 November 2022, which focused on the occurrence rate of neonatal hypothermia and its influencing factors. The meta-analysis revealed a neonatal hypothermia occurrence rate of 52.5% (95% CI: 0.37, 0.68) (Figure 2), which was lower than the findings of Yu et al.²⁹ and Mank et al.,³⁰ but higher than those of Girma et al.¹⁶ and Nguyen et al.,³² which may be caused by confounding factors such as sample size, survey area and study population.

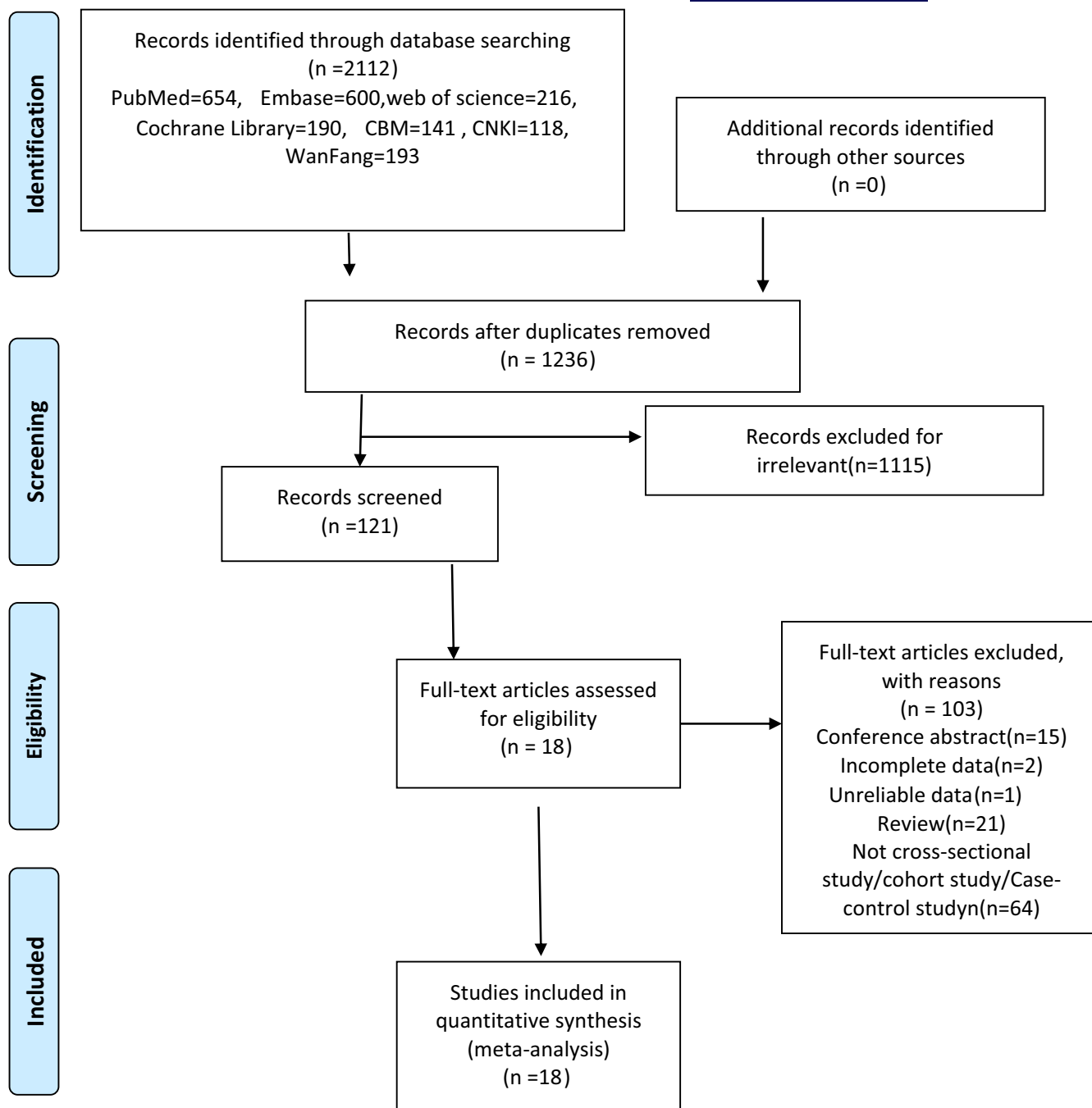


FIGURE 1 Flow diagram of the study selection.

In our study, eight factors were found to be closely related to neonatal hypothermia. Lack of skin-to-skin contact can easily lead to hypothermia. The World Health Organisation (WHO) emphasises the importance of early basic health care for newborns. It recommends that maximum skin-to-skin contact with the mother should be initiated within 10 min after birth and should last for more than 90 min. This practice can help stabilise a newborn's body temperature and reduce the risk of hypothermia.^{35,36} Most studies have confirmed this point.^{37–39} Therefore, hospitals should actively promote maternal and infant skin-to-skin contact and enhance awareness and training of medical staff in this practice. We also found

that newborns who were not wearing a cap were more likely to develop hypothermia. This is mainly because of the larger head size of newborns, open fontanelles and suture lines.⁴⁰ Belghazi et al.⁴¹ found that 50% of newborns lose heat through their head. Similarly, McCall et al.² discovered that hats lined with polyethylene can reduce heat loss in premature infants by 63%–72%. Therefore, timely wearing of caps and provision of warm materials are recommended for newborns.

In our study, we found that delayed breastfeeding increases the risk of hypothermia. Timely breastfeeding can provide the energy needed after birth and reduce the occurrence of neonatal

TABLE 1 Characteristics of the included studies.

No.	Author	Year	Country	Study types	Neonatal hypothermia definition	Object	Total sample size	Incidence rate	Risk factors	Quality assessment score
1	Alebachew Bayih W et al.	2019	Ethiopia	Cross-sectional study	Newborn's body temperature is below 36.5°C	All newborns that were within 6 h of their delivery	403	66.3%	1, 2, 3, 4, 5, 6	9, High
2	Delavar M et al.	2014	Iran	Cross-sectional study	Axillary temperature <36.0°C	Temperatures were recorded immediately after birth and at the age of 1, 2 and 4 h after birth	522	86.6%	8	9, High
3	Demissie BW et al.	2018	Ethiopia	Cross-sectional study	Axillary temperature <36.5°C	Neonates	356	64%	5, 31, 1, 9, 11	8, High
4	Deng et al.	2022	China	Case-control study	Hypothermia <36.5°C	Newborns born within 24 h after delivery	202	12.8%	28, 7, 29	7, High
5	Girma B et al.	2021	Ethiopia	Case-control study	Axillary temperature <36.50°C	Neonates admitted to NICUs within 28 days of delivery	219	33.3%	9, 10, 11, 1, 12, 2	7, High
6	Mank A et al.	2016	Netherlands	Case-control study	Hypothermia was defined as <36.5°C	Hypothermia in Preterm Infants in the First Hours after Birth	80	93.0%	13	7, High
7	Merazzi D et al.	2020	Italy	Cross-sectional study	According to the WHO as temperature <36.5°C	Inborn healthy term or late preterm infants during the immediate postpartum period and before the nursery room entry	4722	59.6% in 2016, 48.4% in 2017	14, 15, 1, 16, 17	7, Medium
8	Miller SS et al.	2011	USA	Cross-sectional study	According to the WHO as temperature <36.5°C	Very low birth weight (VLBW) infants	8782	56.2%	10, 7, 18, 11, 19, 20	8, High
9	Mukunya D et al.	2021	Uganda	Cross-sectional study	Axillary temperature was <36.5°C	Neonates	1330	51.0%	21, 10, 9	10, High
10	Mullany LC et al.	2010	Nepal	Cohort study	Axillary measures of <35.0°C	Neonates	23662	10.0%	22, 10, 5, 9, 16	9, High

TABLE 1 (Continued)

No.	Author	Year	Country	Study types	Neonatal hypothermia definition	Object	Total sample size	Incidence rate	Risk factors	Quality assessment score
11	Ng'eny JC et al.	2020	South Africa	Case-control study	Hypothermia as a skin temperature <36.5	At 24h of life in very low birth weight infants (VLBW)	453	Admission hypothermia was 46.1%, with 38% developing hypothermia within 24-h following admission	10, 8, 11	7, High
12	Nguyen L et al.	2022	Canada	Case-control study	Hypothermia as a skin temperature <36.5	Infants born at ≥34 weeks' gestation	440	26.8%	5	7, High
13	Nyandiko WM et al.	2021	Kenya	Cohort study	An axillary temperature below 36.5°C	Admission to NICU	372	73.7%	1, 5, 10, 23, 2, 24	7, High
14	Ogunlesi TA et al.	2009	Nigerian	Cross-sectional study	Babies with skin (axillary) temperature <36.5°C were considered hypothermic	Admission in Nigerian babies <72 h of age	111	67.6%	5, 9	7, Medium
15	Tasew H et al.	2018	Ethiopia	Case-control study	Hypothermia <36.5°C	Admitted to the neonatal intensive care unit	264	33.3%	9, 10, 5, 18, 1, 12, 25	7, High
16	Ukke GG et al.	2019	Ethiopia	Cross-sectional study	Axillary temperature <36.5°C	Neonates admission to neonatal intensive care units	346	50.3%	10, 9, 25, 30, 4	7, Medium
17	Urubuto F et al.	2021	Rwanda	Cross-sectional study	Hypothermia <36.5°C	Admission at a tertiary neonatal unit	1021	27.0%	5, 8, 26	10, High
18	Yu YH et al.	2020	China	Cohort study	Axillary temperature of <36.5°C	Preterm infants with birth weights (BW) <1500 g and gestational ages (GAs) <34 weeks born in 28 NICUs	1247	88.2%	7, 27, 11, 18	8, High

Note: 1. No skin-to-skin contact; 2. Not wearing a cap; 3. No warm intra-facility transportation; 4. Mothers having obstetric complication; 5. Prematurity; 6. Neonatal health problem; 7. Caesarean section; 8. Natural labour; 9. Delayed breastfeeding; 10. Low birth weight; 11. Asphyxia resuscitation after birth; 12. Night-time delivery; 13. Hypothermia at admission; 14. Analgesia administration; 15. Delayed cord clamping (>60" vs. <60"); 16. Maternal temperature; 17. Room temperature; 18. Low APGAR score; 19. Multiple gestation; 20. Race (Black); 21. Home birth; 22. Sex (Female); 23. No thermal education; 24. No at least 1KMC step; 25. Bathed within 24h; 26. Place of birth (out); 27. Antenatal use of full course of steroid; 28. Maternal age; 29. Neonatal diseases occurred within 24 h after birth; 30. Admissions during cold season; 31. Age of Neonate <24h.

(Continues)

TABLE 2 Meta-analysis and sensitivity analysis of the risk factors for neonatal hypothermia.

No.	Risk factors	Number of included studies	Heterogeneity test		Meta-analysis		Sensitivity analysis	
			I^2	p	Model of effect	OR (95% CI)	p	OR (95% CI)
1	No skin-to-skin contact	6	70	0.005	Random model	2.77 (1.92, 3.99)	<0.001	2.12 (1.81, 2.49)
2	Prematurity	8	89	<0.001	Random model	2.51 (1.77, 3.55)	<0.001	1.43 (1.32, 1.55)
3	Caesarean section	3	83	0.002	Random model	1.86 (1.01, 3.45)	0.05	1.35 (1.16, 1.58)
4	Delayed breastfeeding	7	86	<0.001	Random model	2.26 (1.54, 3.31)	<0.001	1.27 (1.17, 1.38)
5	Low birth weight	8	72	<0.001	Random model	1.73 (1.45, 2.07)	<0.001	1.54 (1.44, 1.65)
6	Asphyxia resuscitation after birth	5	76	0.002	Random model	2.22 (1.34, 3.68)	0.002	1.36 (1.21, 1.54)
7	Low APGAR score	3	74	0.02	Random model	1.97 (1.05, 3.70)	0.03	1.35 (1.20, 1.15)
8	Not wearing a cap	3	0	0.54	Fixed model	1.81 (1.58, 2.08)	<0.001	1.81 (1.58, 2.08)

hypoglycaemia and hypothermia. Therefore, it is also recommended that neonatal breastfeeding should be initiated within 30–60 min after delivery, and breast milk should be fed every hour in the first 24 h after birth.⁴²

Newborns with very low birth weight (VLBW) and extremely low birth weight (ELBW) are more prone to develop hypothermia, as supported by other studies.^{7,43} This susceptibility is primarily attributed to insufficient brown adipose tissue production, limited energy reserves, and poor sucking and swallowing reflexes. Infants weighing 2000–2499 g and 1500–1999 g had a 1.49–4.32 times higher risk of hypothermia, and the incidence of hypothermia increased with weight loss.²⁹ It is worth noting that Ogunna et al.⁴⁴ found that for transported low-birth-weight infants, the incidence of hypothermia was as high as 36%–47%, which was related to prolonged ventilation time and increased mortality before neonatal nursing discharge. Therefore, low birth weight, as a population with a high incidence of hypothermia, requires extra vigilance and more detailed temperature management strategies.

In our study, low Apgar scores and resuscitation for asphyxia were identified as factors influencing neonatal hypothermia. When the oxygen supply to the brain of a newborn is insufficient owing to asphyxia, it can lead to damage to the brain tissue and affect the central regulation of the newborn's body temperature.^{24,29} One study found that 86% of newborns were exposed to temperatures <25°C during resuscitation.²³ The longer a newborn is exposed to these temperatures, the more heat that is lost, which increases the risk of hypothermia. Therefore, it is necessary to take thermal measures for newborns who experienced suffocation and require resuscitation.

We also found that mode of delivery was a risk factor for neonatal hypothermia. One study from the USA²⁴ and two studies from China^{29,34} showed that caesarean section increases the risk of hypothermia, possibly because of cold operating room temperatures. While the WHO recommends operating room temperatures between 25 and 28°C, excessive temperatures can increase discomfort for doctors. Consequently, achieving the recommended temperatures in most operating rooms remains challenging.¹⁰ During a caesarean section, the use of narcotic drugs may affect the brain's thermoregulatory centre, resulting in a decrease in the newborn's body temperature. Therefore, during a caesarean section, it is crucial to enhance the ambient temperature and implement measures to ensure the newborn stays warm. Owing to the limited number of included studies, we were unable to conduct a meta-analysis with fewer than three studies. However, it should be noted that in economically disadvantaged areas of Africa,³¹ due to inadequate medical resources and poor childbirth conditions, natural childbirth may also result in hypothermia, natural births occurred in an open hall with ambient temperatures below 25°C, and the availability of staff was limited, resulting in less attention being given to the newborns during natural childbirth compared to those born via caesarean section. Therefore, it is necessary to enhance the natural childbirth environment in these areas and allocate more human resources to decrease the incidence of neonatal hypothermia.

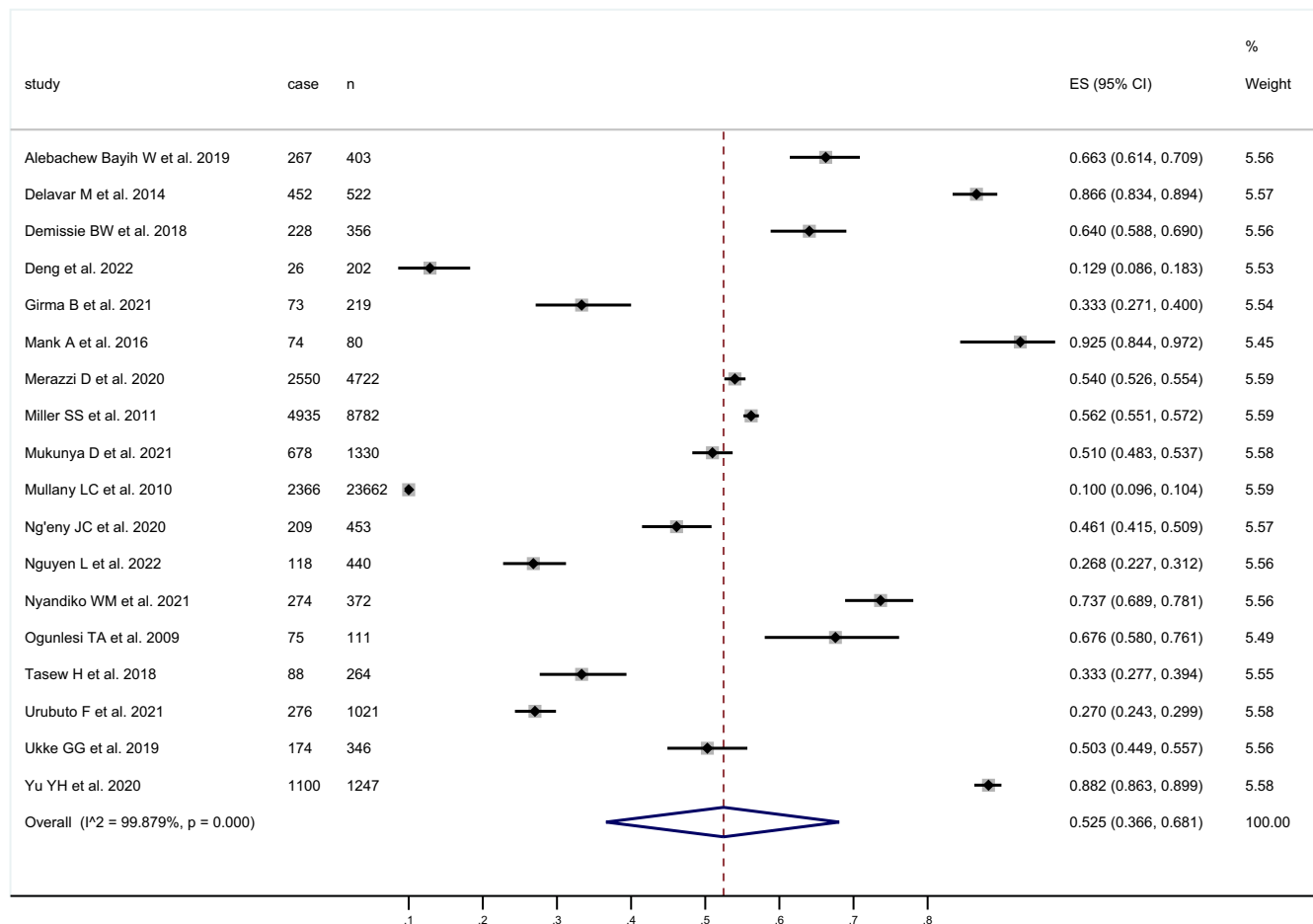


FIGURE 2 Forest map of incidence of neonatal hypothermia.

4.1 | Limitations

This study has the following limitations: (1) Due to the limited number of included articles, only specific influencing factors were analysed. Furthermore, a subgroup analysis of potential factors, such as developed and developing countries, different economic levels, medical facilities, and knowledge of heat preservation among medical staff and mothers, was not conducted to verify specific conclusions. (2) This study focused solely on Chinese and English literature, excluding literature published in other languages and grey literature. Studies included in the analysis exhibited a certain degree of publication bias. Therefore, future research should include multicentre, large-sample cross-sectional, cohort, case-control and related qualitative studies to enhance the effect size and further investigate the incidence and risk factors of neonatal hypothermia.

5 | CONCLUSION

In this meta-analysis, we found that several factors are associated with an increased risk of neonatal hypothermia. These factors included no skin-to-skin contact, not wearing a cap, Prematurity, low

birth weight, delayed breastfeeding, low Apgar scores, asphyxia requiring resuscitation after birth and caesarean section. Medical staff can refer to the results of this study to enhance the assessment and screening of neonatal hypothermia, enabling prompt and effective interventions to minimise the occurrence of low birth weight.

AUTHOR CONTRIBUTIONS

Jing Ruan: Conceptualization; methodology; project administration; writing – original draft; data curation. **Xuemei Zhong:** Data curation; conceptualization; methodology; writing – original draft; project administration. **Lijiao Qin:** Software; data curation; formal analysis; resources. **Jiaxuan Mai:** Software; formal analysis; data curation; resources. **Jiaying Chen:** Writing – review and editing; validation; supervision. **Huiyang Ding:** Writing – review and editing; supervision; validation.

ACKNOWLEDGEMENTS

We would like to express our gratitude to all the authors who contributed to this work and to Professor Deqing Cai for his guidance in the literature search.

FUNDING INFORMATION

This study did not receive financial support.

CONFLICT OF INTEREST STATEMENT

The authors declare that they have no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data sets used and analysed during the current study are available from the corresponding author upon reasonable request.

ETHICS APPROVAL AND CONSENT TO PARTICIPATE

Not applicable.

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How to cite this article: Ruan J, Zhong X, Qin L, Mai J, Chen J, Ding H. Incidence and risk factors of neonatal hypothermia: A systematic review and meta-analysis. *Acta Paediatr*. 2024;113:1496–1505. <https://doi.org/10.1111/apa.17249>